
Exercise #PT-F2: *Model Comparison*

Objective: To implement and fine-tune a probabilistic model (Multinomial Logistic Regression) and a geometric model (Linear SVM) using scikit-learn to classify wine cultivars.

Exercise Description:

1. Data Preparation & Transformation:

Use the `StandardScaler` to ensure the chemical features are normalized—this is a requirement for SVM

- **Dataset:** `sklearn.datasets.load_wine()` (3 classes, 13 chemical features).
- **Partitioning:** 80% Training, 20% Testing.
 - The 80% is your "Training Set" for k-Fold.
 - The 20% is your Test data.
- **The Pipeline:** Use `Pipeline` from `sklearn.pipeline` to combine scaling and modeling into one step, preventing data leakage.

2. Experimental Design (The Fine-Tuning Process):

Both models must be optimized using `GridSearchCV` with 10-fold Cross-Validation

Model A: Multinomial Logistic Regression (MLR)

- **Logic:** Uses the **Softmax** function to provide class probabilities.
- **Fine Tuning Steps:**
 1. Set `multi_class='multinomial'` and use the `lbfgs` solver.
 2. **Fine-Tune Hyperparameter C**
 3. **Task:** Test as many C values as you can and prove through your results which value generates the best performance.
 4. **Additional Task.** Fine Tune other parameters and determine if it will still improve the top model performance.

Model B: Linear Support Vector Machine (Linear SVM)

- **Logic:** Searches for the **Maximum Margin Hyperplane** to separate classes geometrically.
- **Fine Tuning Steps:**
 1. Use `LinearSVC()` or `SVC(kernel='linear')`.
 2. **Fine-Tune Hyperparameter C:**
 3. **Task:** Test as many C values as you can and prove through your results which value generates the best performance.
 4. **Additional Task.** Fine Tune other parameters and determine if it will still improve the top model performance.

3. Evaluation Metrics

- a. Cross-Validation Results: Report the Mean Accuracy and Standard Deviation across the 10 folds.

- b. Generate a confusion matrix using the 20% Final Testing Set. Sample command is as shown:
 - c. From the confusion matrix, extract and report Precision, Recall (Sensitivity), and the F1-Score.
4. IEEE Report Structure:

Section I: Introduction: Define multiclass classification and the goal of chemical profiling.

Section II: Methodology: Document the Python library versions, the 80/20 split, and the use of `StandardScaler`.

Section III: Experimental Design:

III-A: Discuss MLR and the effect of the Regularization parameter.

III-B: Explain the SVM geometric approach and the impact of the Cost (C) parameter on the margin.

Section IV: Comparative Results:

Table 1: Hyperparameter Optimization. Show the "Best C " for both models.

Table 2: Final Test Metrics. Compare Accuracy, Kappa, and F1-score.

Figure 1: Importance Comparison. Side-by-side bar charts of feature weights.

Section V: Conclusion: Summarize whether the probabilistic or geometric approach was more robust for this chemical data.
5. Deliverables
 - i. **IEEE Formatted Report (PDF):** A technical analysis of the comparison of the performance of the two models.
 - ii. **Supporting Python Files:** A clean `.ipynb` notebook using `scikit-learn` pipelines.
 - iii. **Updated Experiment Log (.xlsx):** Log of all tested C values and their corresponding Mean CV Accuracy along with other hyperparameters.

Tip: When using `LinearSVC` in Python, if the model fails to converge during tuning, increase the `max_iter` parameter to 10000.
6. Submit your report along with the all supporting files and `xlsx` files of your experiment in the designated folder.